CODE FILE

Password Strength Analyzer

Author: Harishankar AS

Python source code (.py equivalent)

Note: Use only made-up test passwords. Do not enter a real account password.

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"""
Project: Password Strength Analyzer
Author: Harishankar AS

Educational cybersecurity project. Checks a test password locally and
provides basic improvement suggestions. It does not save or transmit it.
"""

import re
import getpass


def analyze_password(password):
    """Return a score, strength label, and improvement suggestions."""
    score = 0
    suggestions = []

    if len(password) >= 8:
        score += 1
    else:
        suggestions.append("Use at least 8 characters.")

    if len(password) >= 12:
        score += 1
    else:
        suggestions.append("Consider using 12 or more characters.")

    if re.search(r"[A-Z]", password):
        score += 1
    else:
        suggestions.append("Add at least one uppercase letter (A-Z).")

    if re.search(r"[a-z]", password):
        score += 1
    else:
        suggestions.append("Add at least one lowercase letter (a-z).")

    if re.search(r"\d", password):
        score += 1
    else:
        suggestions.append("Add at least one number (0-9).")

    if re.search(r"[^A-Za-z0-9]", password):
        score += 1
    else:
        suggestions.append("Add a special character (for example, ! or #).")

    common_passwords = {
        "password", "password123", "12345678",
        "qwerty123", "admin123", "letmein"
    }
    if password.lower() in common_passwords:
        score = min(score, 1)
        suggestions.append("Avoid common or easily guessed passwords.")

    if score <= 2:
        strength = "Weak"
    elif score <= 4:
        strength = "Moderate"
    else:
        strength = "Strong"

    return score, strength, suggestions


def main():
    print("=" * 45)
    print("       PASSWORD STRENGTH ANALYZER")
    print("       Project by Harishankar AS")
    print("=" * 45)
    print("Your password is checked locally and is not saved.")
    password = getpass.getpass("Enter a made-up test password: ")
    score, strength, suggestions = analyze_password(password)
```

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    print(f"\nStrength: {strength} ({score}/6 checks passed)")

    if suggestions:
        print("\nSuggestions:")
        for item in suggestions:
            print(f"- {item}")
    else:
        print("Good job! Your password passed all basic checks.")

    print("\nSecurity tip: Use a unique passphrase for every account and")
    print("enable multi-factor authentication whenever possible.")


if __name__ == "__main__":
    main()
```